Documents for Go2Joy · 2026-10-24

SharpTalk for Go2Joy

`HA sign-in integration spec v1.4`

Sign-in integration spec v1.4 for the Go2Joy HA dev team (2026-10-24). Done means every item in the checklist at the top passes. Schedule and owners: work request. For printing or sending: PDF

> When an HA staff member opens the widget while signed in, **the HA server signs the staff and hotel data**, and HA FE passes that package to the widget **unchanged**. SharpTalk verifies the signature, registers the hotel as a business customer and the staff member as a contact, continues conversations per hotel account, and routes to the team for the hotel's region. **In v1.4, HA does not calculate the region.** HA sends only the province number (`province_sn`). SharpTalk sets the name and region from the Go2Joy province list.

Item	Details
To	Go2Joy Hotel Admin (HA) dev team (backend, FE) · Go2Joy PM
From	SharpTalk (Amoeba)
Date / version	2026-10-24 · **v1.4** (v1.3 2026-10-22 · v1.0 PDF 2026-10-20 are obsolete)
Environments	HA UAT `https://go2joy-ha-uat.go2joy.io` ↔ SharpTalk staging `https://shoptalk.amoeba.site` · Production is the Go2Joy server `https://sharptalk.go2joy.vn` (install pending)
Related docs	Work request (WR numbers, schedule) · v1.3 (previous spec)
Sample code	sharptalk_identity.php · vectors.php (runs the test vectors)

✅ What HA must do — checklist

This checklist is the core of this document. An item is done when its check passes.

A. HA backend (WR-01 · WR-02) — Tue 10/27

- ☐ **A1. Keep the signing key (embed secret) only in HA server environment variables.** Do not put it in FE, the browser, the app, logs or documents. SharpTalk sends the staging value over a secure channel (on the day WR-02 is decided). Check: the secret string appears nowhere in the repository or bundles.
- ☐ **A2. Create an endpoint that can be called only with a signed-in session** (e.g. `GET /api/v1/staff/sharptalkIdentity`). Check: a call without sign-in returns 401.
- ☐ **A3. Read values from the HA session and DB.** Do not build them from values sent by the browser.
- ☐ **A4.** `userId = "ha:" + userSn`**.** Use the staff number, which never changes. Do not use the login ID (`go2joysupport` etc.).
- ☐ **A5. Exactly 4 attributes**: `hotel_id` · `hotel_name` · `province_sn` · `role`.
  - **Do not send keys outside this list**, such as `hotel_code` · `hotel_group_id` · `user_id` · `user_sn` · `district` (they are left out of the signature calculation, so you get 401).
  - Do not send `region` · `province` (name) either. During the transition period this is not 401, but `province_sn` takes priority. After the transition it becomes 401.
- ☐ **A6.** `province_sn` **= the Go2Joy** `provinceSn` **of the selected hotel** (Province Feed `sn`, numeric string, e.g. Hồ Chí Minh = `"1"`).
- ☐ **A7.** `role` **= the HA role code** (a value that does not depend on the UI language). Send SharpTalk the **code table** (code · Vietnamese/English display name).
- ☐ **A8. When "All hotels" is selected**, leave out `hotel_id` · `hotel_name` · `province_sn` and sign only `role`.
- ☐ **A9. Follow the normalization order**: **NFC → collapse whitespace (incl. NBSP) → trim → 120 chars (**`mb_substr`**)**. Requires the PHP `intl` · `mbstring` extensions.
- ☐ **A10. Leave out empty values entirely (drop the field)**. This applies to attributes and to `name` · `email` · `phone` (an empty `email` returns 400).
- ☐ **A11.** `hash = hex(HMAC-SHA256(secret, signing string))`**.** The signing string uses the format in 5.1 (keys in alphabetical order).

- [ ] **A12. Test vectors T1 ~ T6 all match** (5.4). Send the results to SharpTalk.
- [ ] **A13. Return the response package over HTTPS only to the signed-in session. Do not log it.**

B. HA FE (WR-04) — Wed 10/28

- [ ] **B1. Do not build the package in the browser. Delete** the current `sharptalkSignIn` (built from profile values) and the initial value `hash: "testing"`.
- [ ] **B2. Take the A2 response and pass it to `SharpTalk.identify(response)` unchanged** (do not add or change keys).
- [ ] **B3. Call again on**: sign-in confirmed · page reload · **hotel selection change (incl. All hotels)** · HA language change.
- [ ] **B4. Call `SharpTalk.logout()` on**: sign-out button · **automatic sign-out on session/token expiry** · right before another account signs in · sign-out in another tab (`storage` event etc.).
- [ ] **B5. `locale`**: same as now, `localStorage.lang` → `vi` / `en`. Call `setLocale` when it changes.
- [ ] **B6. Allow the SharpTalk domain in CSP (`script-src` · `frame-src` · `connect-src`) and CORS.**
- [ ] **B7. Put the 3 SharpTalk domain uses** (`embed.js` src · `widgetUrl` · `ivy:event` origin check) **behind one environment setting** (so you change it once when moving to production). Check (all of B): DevTools › Network `POST …/public/embed/identify` **200** (`consentState: "granted"`), and the page receives `ivy:event identified {ok: true}`.

C. Joint UAT check (WR-05) — 10/29 ~ 10/30

- [ ] **C1. All items in the section 8 checklist pass** (together with SharpTalk).
- [ ] **C2. Test accounts**: 1 `Quản lý` (multiple hotels) + 1 `Lễ tân` (1 hotel), including hotels in the north and the south (WR-14).

D. Production deploy (WR-06) — before production UAT

- [ ] **D1. Apply the production embed secret.** It differs from staging (SharpTalk sends it over a secure channel).

- ☐ **D2. Switch to the production domain**: `https://sharptalk.go2joy.vn/widget/embed.js` · `widgetUrl` · origin check.
- ☐ **D3. Apply production CSP and CORS. Share the HA deploy schedule in advance.** Check: identify returns 200 on production HA. A package made with the staging secret returns 401 on production.

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v1.4 changes

Item	v1.0 (10/20 PDF)	v1.3	v1.4
`userId`	Login ID	`"ha:" + userSn`	Same as v1.3
Signed attribute keys	`hotel_id` · `hotel_name` · `province` · `region` · `role`	Same	**`hotel_id`** · **`hotel_name`** · **`province_sn`** · **`role`**
Province	Province name	Province name	**`province_sn`** = Go2Joy `provinceSn`
Region	Calculated by HA (3 values)	Calculated by HA (5 codes)	**Not sent**. Calculated by SharpTalk
Role	Display name (`Lễ tân`)	Display name	**Role code**
Normalization	trim · 120 chars	+ NFC	**NFC → collapse whitespace → trim → 120 chars**
"All hotels"	Not defined	Call again only	Drop the 3 hotel keys, sign `role` only
Empty values	Empty attributes · email left out	Same	Empty `name` · `email` · `phone` also left out
Test vectors	T1 · T2	T1 ~ T5	**T1 ~ T6** (previous vectors no longer used)

Why the change

- If both HA and SharpTalk keep a region table, they drift apart. The single source is the Province Feed of the Go2Joy Partner API.
- Province names and role display names change with the HA language (vi/en), so the signed values could differ. Numbers and codes do not change.

Transition period: For now, SharpTalk still accepts the v1.3 format (`province` · `region` keys). If `province_sn` is also sent, **the values set from `province_sn` take**

priority. When HA has moved to v1.4 and passed UAT, we will end support for the v1.3 keys and announce the date.

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1. What SharpTalk does

Step	Details
Verify	Builds the signing string from the received values as-is and compares the HMAC. On mismatch: 401 E5048 (stays guest)
Province · region	`province_sn` → fills **province name and region** from the Go2Joy province list (80 entries, Partner API Province Feed, refreshed daily). A number not in the list is saved without a region and goes to the default assignee
Register	Hotel = business customer (`hotel_id`), staff = contact (`userId`), role recorded
Continuity	Continues previous conversations per hotel account (after session end, on another PC, with another staff member)
Routing	Consult category chosen by the customer × hotel region → team × work label (rules managed by Go2Joy in the console)

Region table (Province Feed `region`)

No.	Code	Name
1	`south`	South
2	`north`	North
3	`central`	Central
4	`north_center`	North Central
5	`south_center`	South Center
10	`region_10`	Not documented (17 cities of Manila). To be fixed after the reply to work request WR-27

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2. Data sent (v1.4)

Field	Signed	Value	HA source (example)	Example
`userId`	Yes	`"ha:" + userSn`	Staff number in HA DB	`ha:1001`
`attributes.hotel_id`	Yes	Selected hotel `hotelSn` (string)	Selected hotel in the session	`"1721"`

Field	Signed	Value	HA source (example)	Example
`attributes.hotel_name`	Yes	Hotel name	Hotel in HA DB	`A In Hotel Del Luna`
`attributes.province_sn`	Yes	`provinceSn` of the hotel's province (string)	Hotel in HA DB → provinceSn	`"1"`
`attributes.role`	Yes	HA role code	Role key in HA DB	`RECEPTIONIST` (example. Real codes come with the A7 reply)
`name` · `email` · `phone`	No	For display. Drop the field when empty	Profile	`G2J SUPPORT`

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3. Request flow

```
HA FE ──(signed-in session)──▶ HA server  GET /api/v1/staff/sharptalkIdentity
                          └ values from session/DB → normalize → sign → return packa
ge
HA FE ── SharpTalk.identify(package unchanged) ──▶ widget iframe
widget ── POST /api/v1/public/embed/identify ──▶ SharpTalk
                          └ verify signature → province_sn → province name/region →
business customer/contact → link conversation
widget ── ivy:event { event: "identified", ok } ──▶ HA FE (optional: log)
```

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4. Signing string

```
no attributes   : userId
with attributes : userId + "\n" + key=value + "\n" + key=value ...   (keys in alpha
betical order)

hash = hex( HMAC-SHA256( key = SHARPTALK_EMBED_SECRET, message = signing string (UT
F-8) ) )
       → 64 lowercase hex chars

Example (T2):
ha:1001
hotel_id=1721
hotel_name=A In Hotel Del Luna
province_sn=1
role=RECEPTIONIST
```

- Key order: `hotel_id` → `hotel_name` → `province_sn` → `role`.

- The values you send must be **exactly the same** as the values you signed (including spaces, letter case and tone marks).

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5. Implementation

5.1 HA server (PHP) — sharptalk_identity.php

```
<?php
// SharpTalk identify package — HA server (spec v1.4). Requires PHP intl + mbstrin
g.

/** NFC → collapse whitespace (incl. NBSP) → trim → 120 chars. Order matters. */
function sharptalk_clean($value): string {
    $v = Normalizer::normalize((string) $value, Normalizer::FORM_C);
    if ($v === false) { $v = (string) $value; }
    $v = preg_replace('/[\s\x{00A0}\x{2000}-\x{200B}\x{3000}]+/u', ' ', $v);
    $v = trim($v);
    return mb_substr($v, 0, 120, 'UTF-8');
}

/**
 * @param string     $userSn   staff number from the HA DB (never the login ID)
 * @param array|null $hotel    ['sn' => …, 'name' => …, 'provinceSn' => …] or null
for "all hotels"
 * @param string     $roleCode HA role code (language independent)
 * @param array      $profile  ['name' => …, 'email' => …, 'phone' => …] — display
only, not signed
 */
function sharptalk_identity(string $userSn, ?array $hotel, string $roleCode, array
$profile, string $secret): array {
    $attributes = ['role' => $roleCode];
    if ($hotel !== null) {
        $attributes['hotel_id']    = $hotel['sn'];
        $attributes['hotel_name']  = $hotel['name'];
        $attributes['province_sn'] = $hotel['provinceSn'];
    }
    $clean = [];
    foreach ($attributes as $k => $v) {
        $c = sharptalk_clean($v);
        if ($c !== '') { $clean[$k] = $c; }          // empty values are left out o
f signing AND payload
    }
    ksort($clean, SORT_STRING);
    $userId = 'ha:' . sharptalk_clean($userSn);
    $lines = [$userId];
    foreach ($clean as $k => $v) { $lines[] = $k . '=' . $v; }
    $hash = hash_hmac('sha256', implode("\n", $lines), $secret);
```

```
    $package = ['userId' ⇒ $userId, 'hash' ⇒ $hash];
    foreach (['name', 'email', 'phone'] as $f) {     // unsigned; drop when empty
        $c = sharptalk_clean($profile[$f] ?? '');
        if ($c ≠ '') { $package[$f] = $c; }
    }
    if ($clean) { $package['attributes'] = $clean; }
    return $package;
}

// Controller (login session required):
// $staff = currentStaff(); $hotel = selectedHotelOrNull();
// return json(sharptalk_identity($staff→sn, $hotel ? ['sn' ⇒ $hotel→sn, 'name'
⇒ $hotel→name,
//        'provinceSn' ⇒ $hotel→provinceSn] : null, $staff→roleCode,
//        ['name' ⇒ $staff→fullName, 'email' ⇒ $staff→email, 'phone' ⇒ $staff-
>mobile],
//        getenv('SHARPTALK_EMBED_SECRET')));
```

5.2 HA FE

```
<script>
  window.SHARPTALK_WIDGET_CONFIG = {
    shop: "https://www.go2joy.vn",
    widgetUrl: SHARPTALK_ORIGIN + "/widget",        // B7: one setting per environm
ent
    agent: "hotel-partner",
    locale: (localStorage.getItem("lang") || "vi").toLowerCase()
    // no `user` block, no hash here (B1)
  };
  window.SharpTalk = window.SharpTalk || { q: [] };
  function sharptalkCall(method, arg) {
    if (typeof window.SharpTalk[method] ≡ "function") window.SharpTalk[method](ar
g);
    else window.SharpTalk.q.push([method, arg]);
  }
</script>
<script src="SHARPTALK_ORIGIN/widget/embed.js" defer></script>
```

```
// B2 · B3: after sign-in, on reload, on hotel change (incl. "all hotels"), on lang
uage change
async function sharptalkSignIn() {
  const pkg = await haApi.get('/api/v1/staff/sharptalkIdentity'); // signed on the
server
  sharptalkCall('identify', pkg);                                  // UNCHANGED
}
// B4: sign-out button, session/token expiry, before another account signs in, othe
r tab signed out
function sharptalkSignOut() { sharptalkCall('logout'); }
window.addEventListener('storage', (e) ⇒ { if (e.key ≡ 'profile' && !e.newValue)
sharptalkSignOut(); });
```

```
// B5
function sharptalkSetLocale(lang) { sharptalkCall('setLocale', String(lang || 'v
i').toLowerCase()); }
// optional log
window.addEventListener('message', (e) => {
  if (e.origin !== SHARPTALK_ORIGIN || e.data?.type !== 'ivy:event') return;
  if (e.data.event === 'identified') console.info('[SharpTalk] identify', e.data.ok
? 'OK' : 'rejected');
});
```

5.3 Package example (T2)

```
{
  "userId": "ha:1001",
  "hash": "fb670d6fbaac555b40b2f9896ec7de0604a1fad83aaf9e892f293f1d3be9faa7",
  "name": "G2J SUPPORT",
  "attributes": { "hotel_id": "1721", "hotel_name": "A In Hotel Del Luna", "provinc
e_sn": "1", "role": "RECEPTIONIST" }
}
```

5.4 Test vectors (v1.4)

The test secret is `test-secret-do-not-use` (not a real secret). The hashes below were computed separately by **the SharpTalk server verification code** and by **the 5.1 PHP sample (PHP 8.3 + intl)**, and they match. Role codes and `userSn` are example values.

#	Input	Expected hash
T1	`ha:1001` · no attributes	`244f6fa4c1c48d077d54bd632b0897042fa413e36d55921517c462e2af51e618`
T2	`ha:1001` · hotel_id= `1721` , hotel_name= `A In Hotel Del Luna` , province_sn= `1` , role= `RECEPTIONIST`	`fb670d6fbaac555b40b2f9896ec7de0604a1fad83aaf9e892f293f1d3be9faa7`
T3	`ha:2002` · hotel_id= `2001` , hotel_name= `Huế Garden Hotel` , province_sn= `32` , role= `OWNER`	`0df74c490621c521a673b7bbd6014e6e3141a086d04c7c2f8de3699d31e581d7`
T4	`ha:3003` · hotel_id= `3001` , hotel_name= `Hạ Long Bay Hotel` , province_sn= `15` , role= `RECEPTIONIST`	`7e5cf87cdd3dc38574ec8f3f9fe6671e9650806d28f4741a55bb508f53b77ea9`
T5	`ha:1001` · role= `MANAGER` (**All hotels**)	`d57dc894632e52a7ca62d97f4cbec924eb33982dff23f57058b7299d24f57f20`
T6	Same values as T3, entered messy: hotel_id= `" 2001"` (leading space), hotel_name= `" Hu" + "ế" + " Garden Hotel "` (decomposed ế · double space · NBSP), province_sn= `32` (number type) → sign after 5.1 normalization	**Must equal T3** `0df74c490621c521a673b7bbd6014e6e3141a086d04c7c2f8de3699d31e581d7`

- Run: `docker run --rm -v "$PWD/scripts/go2joy/php":/w php:8.3-cli sh -c 'apt-get update -qq && apt-get install -y -qq libicu-dev && docker-php-ext-install intl && php /w/vectors.php'`
- Resulting province/region (values SharpTalk saves): T2 Hồ Chí Minh · `south`, T3 Thừa Thiên Huế · `north_center`, T4 Quảng Ninh · `north`, T5 none (default assignee).
- Do not use the v1.0 ~ v1.3 vectors (only T1 is the same).

5.5 SharpTalk responses

Result	Meaning	Action
200	Verified. Session is "verified", business customer and contact linked	—
401 · E5048	Signature mismatch: string · secret · environment mismatch, unknown key, over 120 chars or line break, NFC/whitespace difference, non-numeric `province_sn`	Compare with section 4, rerun 5.4
400	Format error (empty email etc.)	Leave out empty fields
404	No chat session	Widget handles it automatically. Reload

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6. Security

- Keep the secret only in HA server environment variables. Staging and production secrets differ (SharpTalk sends them over a secure channel).
- Anyone holding a signed package can chat as that staff member and see that hotel's previous conversations. Do not log it, and do not skip sign-out handling (B4).
- Whether to add a signature expiry (`ts`) is decided in work request WR-03 (if added, it ships together as v1.5).
- Shared PCs (front desk): automatic sign-out and account switching in B4 are especially important.

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7. SharpTalk status

Item	Status
v1.4 support (`province_sn` → province name/region, v1.3 in parallel)	Staging deploy (target 10/27)
Go2Joy province list, 80 entries (indexed by number)	Loaded on staging · production loaded at install
Temporary unsigned sign-in (for staging UAT)	Ends after A and B are done
Role code display names	After the A7 code table is received

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8. UAT checklist (WR-05)

Check	Expected
Widget while not signed in	Guest, consent notice shown (same as now)
HA sign-in → widget	identify **200**, `identified {ok: true}`, no consent notice, staff name in greeting, "verified" in console
Console › Customers › Business customers	1 hotel (name · province · region, calculated by SharpTalk), staff and role code under contacts
HA language vi ↔ en	Signed values unchanged → same conversation continues, only the reply language changes
Change hotel selection	Business customer and conversation switch to the new hotel
"All hotels" → connect to agent	200, no hotel, default assignee
Connect to agent from hotels in each region (north · north_center · south_center · south) → choose consult category	Team routing follows console rules, assignee shown in alarm center
Hotel name with many tone marks (incl. decomposed input)	200
Staff with empty email	200 without the email field
Sign-out · session expiry · another account signs in	Guest (or the new account), previous staff conversations not visible
Change one signed value without re-signing	401, guest. `name` · `email` · `phone` are not signed, so changing them still gives 200
Add a key outside the list (e.g. `district`) and sign	401 (unknown keys are left out of the signature calculation)